

# Reviewer Pack — Erdős-Ko-Rado Shadow Defect Lower-Bounds Disjointness Discre...

Entry 78c33f7389a9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-04-27 04:23:28 UTC

## Erdős-Ko-Rado Shadow Defect Lower-Bounds Disjointness Discrepancy

**Entry ID:** 78c33f7389a9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-04-27 04:23:28 UTC

### 1. Conjecture

**Field A** (mathematical branch): Extremal set theory via Kruskal-Katona shadow profiles and EKR-type intersecting family deficits **Field B** (complexity object): Discrepancy-based randomized two-party communication complexity of partial set-disjointness matrices

#### Statement:

For an  $n \times n$   $\pm 1$  matrix  $M_F$  constructed from a  $k$ -uniform set family  $F \subseteq \{0,1\}^n$  (rows = members of  $F$ , columns = members of  $F$ , entry =  $+1$  if disjoint,  $-1$  if intersecting), let  $s(F)$  be the Kruskal-Katona shadow defect, defined as  $|\partial F| - \text{KK\_min}(|F|, k-1)$  where  $\partial F$  is the  $(k-1)$ -shadow and  $\text{KK\_min}$  is the colex minimum, and let  $\delta(F)$  be the spectral discrepancy  $\|M_F\|_{\infty \rightarrow 1} / (|F|^2)$ . We conjecture that for every  $k$ -uniform  $F$  with  $2 \leq k \leq n/2$ ,  $\delta(F) \geq c \cdot s(F) / (|F| \cdot \text{binomial}(n, k-1))$  for a universal constant  $c \geq 1/8$ , with equality approached only by colex-initial (EKR-extremal) families. A single  $(n, k, F)$  with  $\delta(F) < (1/8) \cdot s(F) / (|F| \cdot C(n, k-1))$  refutes it.

#### Rationale (proposer's reasoning):

Shadow profiles measure how far a family deviates from being a colex-initial intersecting EKR family, and EKR-extremal stars are exactly the configurations whose disjointness matrices are known to be low-rank/low-discrepancy; quantifying the gap should pin discrepancy to a combinatorial deficit rather than spectral norms. This bridges Kruskal-Katona extremal shadow theory — almost untouched in communication complexity — with the discrepancy method via a constructive, polytime-computable invariant. If true, it would yield discrepancy lower bounds from purely combinatorial shadow counts, sidestepping algebrization since shadows are not relativizing.

**Taxonomy category:** COMM\_COMPLEXITY (status at proposal time: alive)

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** dd666fb2883ade96

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

**Acceptance criterion:**

Across all enumerated/sampled  $(n,k,F)$  with  $n \in \{8..16\}$ ,  $k \in \{2,3,4\}$ ,  $\geq 200$  families per  $(n,k)$  plus colex-initial and star families, compute  $R(F) = \delta(F) \cdot |F| \cdot C(n,k-1) / \max(s(F), 1)$ . Conjecture is SUPPORTED iff  $\min R \geq 1/8$  over all tested  $F$ ; FALSIFIED iff any single  $F$  yields  $R < 1/8$ .

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|---------|------------|------------------|------------------|
| RELATIVIZATION | SAFE    | 0.90       | SAFE             | SAFE             |
| NATURAL_PROOFS | SAFE    | 0.90       | SAFE             | SAFE             |
| ALGEBRIZATION  | SAFE    | 0.82       | SAFE             | SAFE             |
| KARP_LIPTON    | SAFE    | 0.95       | SAFE             | SAFE             |

### 4. Novelty audit

**Verdict:** NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - Kruskal-Katona shadow EKR intersecting family discrepancy communication complexity - set disjointness matrix spectral norm discrepancy intersecting family colex - shadow minimum colex initial family disjointness sign matrix discrepancy bound

**Top relevant hits considered:** - [<http://arxiv.org/abs/1809.03967v2>] More Cases Where the Kruskal-Katona Bound is Tight - [<http://arxiv.org/abs/2005.06999v1>] Tight bounds for Katona's shadow intersection theorem - [<http://arxiv.org/abs/2305.10152v1>] On the extremal families for the Kruskal-Katona theorem - [<http://arxiv.org/abs/1408.1376v2>] Factorization Norms and Hereditary Discrepancy - [<http://arxiv.org/abs/1208.6015v1>] The spectral function of a first order elliptic system - [<http://arxiv.org/abs/1009.0997v3>] Spectral asymptotics for Robin problems with a discontinuous coefficient - [<http://arxiv.org/abs/2301.09079v2>] Star discrepancy bounds based on Hilbert space filling curve stratified sampling and its applications - [<http://arxiv.org/abs/1102.4243v2>] Disjointness and unique ergodicity of  $C^*$ -dynamical systems

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only,  $\leq 90$ s wall time per cycle **Execution:** rc=124, elapsed=180.1s

#### 5.1 Generated Python source

```
import random
import math
from itertools import combinations

def binomial(n, k):
    if k > n:
        return 0
    res = 1
    for i in range(k):
        res *= (n - i)
        res //= (i + 1)
    return res

def colex_min(n, k):
    return binomial(n, k)
```

```

def shadow_defect(F, k):
    n = len(F[0])
    s_F = 0
    for comb in combinations(range(n), k-1):
        shadow = [any(all(F[i][j] == F[j][i] for j in range(k)) for i in comb) for _ in range(len(F))]
        s_F += sum(shadow)
    return s_F - colex_min(len(F), k-1)

def spectral_discrepancy(M_F):
    n = len(M_F)
    max_val = 0
    for x_vec in combinations([-1, 1], n):
        for y_vec in combinations([-1, 1], n):
            val = abs(sum(x_vec[i] * M_F[i][j] * y_vec[j] for j in range(n))) / n**2
            if val > max_val:
                max_val = val
    return max_val

def run_trial(seed: int) -> dict:
    random.seed(seed)
    results = []
    for n in [8, 11, 14]:
        for k in [2, 3, 4]:
            for size in range(binomial(n-1, k-1), binomial(n, k)+1):
                F = [tuple(random.sample(range(n), k)) for _ in range(size)]
                s_F = shadow_defect(F, k)
                M_F = [[1 if set(x) & set(y) == set() else -1 for y in F] for x in F]
                delta_F = spectral_discrepancy(M_F)
                R_F = delta_F * size * binomial(n, k-1) / max(s_F, 1)
                results.append((R_F, F))
    min_R, witness_F = min(results)
    conjecture_holds = min_R >= 0.125
    counterexample = "" if conjecture_holds else f"min(R)={min_R}, witness_F={witness_F}"
    return {
        "metric_name": "R(F)",
        "metric_value": min_R,
        "instances_tested": len(results),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37, 53, 71]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result["metric_value"])

    mean = sum(results) / len(results)
    std_dev = math.sqrt(sum((x - mean)**2 for x in results) / len(results))
    support_fraction = sum(1 for r in results if r >= 0.125) / len(results)
    print(f"RESULT: {'SUPPORTED' if support_fraction >= 0.8 else 'FALSIFIED'} mean={mean} std={std_dev}")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

TIMEOUT after 180s

## 8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

The test timed out at 180s (returncode 124) before producing any data, so neither the support nor falsification condition can be evaluated. The critic’s CHALLENGE stands. | next: Reduce the search space (e.g.,  $n \in \{8..12\}$ ,  $k \in \{2,3\}$ , 50 random samples per cell) and optimize the spectral discrepancy computation (use  $\|M\|_{\infty} \rightarrow 1$  via sign-vector enumeration only for small  $|F|$ , or a Grothendieck-style SDP relaxation) to fit within the time budget.

## 11. Audit log (LLM calls)

**Total LLM calls:** 7

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | claude_max    | opus             | 0         | 0   | 19516        |
| 2 | preregistration | claude_max    | opus             | 0         | 0   | 6644         |
| 3 | novelty         | claude_max    | opus             | 0         | 0   | 3795         |
| 4 | novelty         | claude_max    | opus             | 0         | 0   | 9298         |
| 5 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 15050        |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 8591         |
| 7 | judge           | claude_max    | opus             | 0         | 0   | 5162         |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 68056 ms total latency. Provider mix: {'claude\_max': 5, 'ollama\_remote': 2}

(full prompt+response transcripts available in `research/audit/78c33f7389a9.jsonl`)

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/78c33f7389a9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** `research/pvsnp_sandbox/test_*.py` (per-cycle)

**Replay tarball:** `research/replay/78c33f7389a9.tar.gz` (if generated)